

Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch:	Roll No.:
Experiment / assignment / tutorial No. 02	
Grade: AA / AB / BB / BC / CC / CD / DD	
Signature of the Staff In-charge with date	

Title: Source Coding

Objectives:

1. To develop a logic and write a program for implementing source coding algorithm for the given problem statement

OUTCOME: apply information theory in source coding

Problem statement: To encode anyone tongue-twister phrase (given below) using source coding algorithm.

- which wrist watches are swiss wrist watches
- fred fed ted bread and ted fed fred bread
- top chopstick shops stock top chopsticks
- a happy hippo hopped and hiccupped

Procedure:

- Select any one tongue-twister phrase from the given list
- Develop a logic and write a program of source coding algorithm (Huffman coding or LZ coding) to generate a codebook for the symbols in the phrase using MATLAB or any other open source tool.

Expected output:

- Number of different numbers of symbols in the phrase
- Number of code words representing different symbols
- Number of bits needed for coding characters in this phrase using fixed length code
- Theoretical minimum number of bits required to encode the entire phrase assuming that each character is independent of other surrounding character.
- Number of bits needed for encoding the sequence using codebook derived by Huffman/LZ coding algorithm.
- Calculate efficiency of each codebook.

Discussion/Conclusion:

Signature of faculty in-charge

Department of Electronics and Telecommunication Engineering